

Technology Stack

Date: 11 June 2026

Project Name: E-commerce Application

Technology Stack Breakdown

Layer	Technology	Purpose / Justification
Frontend Framework	React.js	Facilitates building dynamic, component-driven Single Page Applications (SPAs) with fast state updates via the Virtual DOM.
Backend Framework	Express.js	A lightweight and flexible Node.js web application framework that provides robust routing and middleware capabilities for RESTful APIs.
Runtime Environment	Node.js	Allows executing JavaScript server-side, enabling full-stack JavaScript development for better team efficiency.
Database	MongoDB	A NoSQL database that pairs perfectly with Node.js and allows flexible schema design, ideal for an evolving E-commerce catalog.
Object Data Modeling (ODM)	Mongoose	Provides a straight-forward, schema-based solution to model application data and relationships within MongoDB.
Authentication	JSON Web Tokens (JWT) & Bcrypt.js	JWT handles stateless session management via secure cookies, while Bcrypt.js encrypts user passwords before saving them to the DB.
Payment Gateways	Payment Gateway (Planned)	Industry-standard APIs for securely handling customer credit card and local payment methods without storing sensitive financial data on the server.
Media Hosting	Cloudinary	Cloud-based image management service used to store, optimize, and serve product photos rapidly via CDN.
Development Tools	Nodemon, Postman	Nodemon restarts the server on file changes; Postman is used for testing API endpoints during development.